

Lesson 02 : Comparison, Logical, & Assignment Operators

Description

Students continue learning basic coding concepts. They create 4 simple projects to learn comparison, logical and assignment operators and 1 bigger project by combining all concepts.

Objectives

Students will:

1. Understand what are comparison operators and use them in a code.
2. Recognize what are logical operators and use them in a code.
3. Explain what are comments and use it in a code.
4. Describe what are assignment operators and use them in a code.
5. Analyze codes and figure out the outcome of a code.
6. Create a project combining all concepts learned during the lesson.

Tools

1. Pycharm

Class Structure

Duration	Topic
5 mins	Preparation
5 mins	Comprehension Check
15 mins	Comparison Operators
15 mins	Logical Operators
10 mins	Assignment Operators
10 mins	Comments
15 mins	Project Combination
10 mins	Review Quiz
5 mins	Wrap Up

Planned Activities	Resources
Preparation (5 mins)	
- Teacher greets students and ask how's their conditions. - Instruct students to open their pycharm. - If possible, teacher can incorporate a mini game during this preparation time to engage with the students.	
Comprehension Check (5 mins)	
- Ask students if they're still remember what they've learned on the previous lesson. - Give students chance to recall the key topics from the previous lesson. - Teacher can ask students regarding particular topic to check students' comprehension. - If possible, teacher can ask students to share their screen and re-explain in brief about materials that were taught on the previous lesson.	
Comparison Operators (15 mins)	
Explain – Comparison Operators - We explain to students that a comparison operator is an operator that can be used to compare between two different data or values. We usually put/store them (data/values) in variables. - As a result from the comparison that was done, the user will get a Boolean result which can be either True or False. - Comparison Operators can be either of the following operators:	File/Code: 1.ComparisonExample.py

- > (Greater Than)
- < (Less Than)
- >= (Greater Than or Equals)
- <= (Less Than or Equals)
- != (Not Equal)
- == (Equals to)

Explain and Show – Comparison Operator Mini Project

- We show and explain the project example to students. So It would be easier for them to understand.
- Tell them in this project they will create a program that will have two different values and print it out to our screen and they should include all of the comparison operators that they have learned from the explanation. Let them explore it by themselves for 5 - 10 min.

Individual Project – Mini Project Creation

- We tell students that now that they know how the project might work, they should try and create it on their own.
- We remind students that in order to create the program successfully, they have to use some of the skills that they have learned during the previous lesson (variables, concatenation, type conversation).
- We give students time to do it by themselves and discuss it together when they are finished.

File/Code:

[2.ComparisonProject.py](#)

Logical Operators (15 mins)

Explain and Show – Logical Operators

- We explain to students that a logical operator is an operator that can help us when we want to check between two different conditions at once.

File/Code:

[3.LogicalExample.py](#)

- Logical Operators can be either of the following operators:
 - **AND** (True if both conditions are true)
 - **OR** (True if one of the condition is true)
 - **NOT** (Reverse the result of an expression)
- Teacher briefly re-explain and gave out examples by writing it on the whiteboard.
- Teacher also shows the project example so that the students better understand how the logical operators work.
- We let students try the logical operators by themselves first.

Explain and Show – Logical Operators Mini Project

- We show students the project and explain that for their next task, we would like them to create a project that will check two different conditions at the same time by using the logical operators that they have learned.
- For example, inside the project example we tried to check whether the name of the favorite food and the amount of food is true when compared to the values that were declared at the beginning.

Individual Activity – Logical Operators

- We ask students to incorporate 2-3 different logical operators to their project to create something similar.
- We can tell students to create it and that they should try to use their own creativity when creating the project.

File/Code:

[4.LogicalProject.py](#)

Assignment Operators (10 mins)

Show & Explain – Assignment Operators

File/Code:

[5.Assignment.py](#)

- We explain that an assignment operator is an operator that can be used to assign values to a variable. For example, the equal sign that we have used to define a variable (For example: `testing = 20`) is one of the examples of an assignment operator.
- We explain that besides the equal sign, there are other assignment operators, such as:
 - **+= (Add AND)**. `x += y` is equivalent to `x = x + y`
 - **-= (Subtract AND)**. `x -= y` is equivalent to `x = x - y`
 - ***= (Multiply AND)**. `x *= y` is equivalent to `x = x * y`
 - **/= (Divide AND)**. `x /= y` is equivalent to `x = x / y`
- We show students the project example and explain how we can use the assignment operator to easily do calculations and assign the new value to our initial variable.

Individual Activity – Guess The Answer Hot Seat

- We tell students to create a new file and ask them to create two variables and add at least 4 assignment operators in it.
- We then tell them that when they are done, we should let students observe other student's code and before running it, we tell them that they should guess the code first.

Comments (10 mins)

Explain – Comments

- We explain to students that a comment is a way for developers to write notes on their code which can be used for different purposes.
- For example:
 - Comments can be used to remind ourselves what a certain part of our code does.
 - Comments can be used to let other people know how our code might work.
 - Comments can be used to disable certain parts of our code.
- There are two types of comments that can be added to our code:

File/Code:

[6.Comments.py](#)

- The first one is called a single line comment, this type of comment is the one that have been mentioned on the video and can be added by adding a # (hashtag) to our code.
- The second one is called a multi line comment and can be used to add comments that span multiple lines. To add Multi line comment we can add `""" """` (3 double quote followed by another 3 double quote to end it).

Explain & Show – Comments Project

- We show students the project and ask them if they still remember what the two types of comments are and how can we add it to our project?
- We then show the project example and re-explain to students what comments do.

Individual Activity – Comment Addition

- We ask students to add comments to the projects that they have created.
- We tell them that on the top, they should add at least their name, the date of the project creation, and the name of the project itself.

Project Combination (15 mins)

Explain & Show - Project Combination

- Teachers show students the project and explain that in the next project, we would like them to combine what they have learned so far.
- Teacher should also explain that inside a statement, we can also add multiple logical statements just like in the example.
- Teachers can ask students why the first statement is true while the second one comes up as false. After we give them time, we can then explain that the first statement has an “and” in the middle which means that it will check whether both statements inside the brackets are True.

File/Code:

[7.Combination.py](#)

- Next, we can see that the statement before the “And” and after the “And” uses an “Or” logical statement (comes up as True if one of the statements is True) which comes up as True for both of them.
- With both statements inside the brackets coming up as True. Combined with the “And” logical statement will then result into a “True”.
- The second print is also similar, but instead of having an “And” logical statement in the middle, we have an “Or” instead.

Individual Activity – Project Creation

- We explain to students that they should now create the project by themselves.
- We remind them that they should add more than one logical operator to their statements and add comments to explain what they are doing.

Review Quiz (10 mins)

Individual Activity – Review Quiz

- We show students the review quiz questions for today and give them some time to answer it by themselves in a notepad file.
- When everyone is done, we tell them to save their answers and review it together.

Wrap Up (5 mins)

Wrapping Up

- We sum up by asking questions, making sure the students remember the most important things.
- Hands up! Who knows what comparison operators and logical operators are?

- We thank students for doing a great job overall.

☐ ☆ Good Luck With The Class !! ☆ ☐